The Moon: Phases, Months and Eclipses

What you will be able to do

1. Explain why the Moon shows phases, and name them in English and in Indian terms.
2. Predict roughly when the Moon rises and sets for any phase.
3. Explain the difference between the sidereal month (27.3 days) and the synodic month (29.5 days).
4. Explain solar and lunar eclipses — and why they do not happen every month (Rahu and Ketu).
5. Recall important eclipses in Indian history and the next ones visible from India.

1. Moon facts worth remembering

Quantity	Value
Average distance from Earth	**3,84,400 km** (about 30 Earth diameters)
Diameter	3,474 km (about $\frac{1}{4}$ of Earth's)
Apparent size in the sky	$\approx 0.5^\circ$ (your little fingernail at arm's length covers it)
Orbit relative to the stars (**sidereal month**)	**27.32 days**
New Moon to New Moon (**synodic month**)	**29.53 days**
Tilt of the orbit to the ecliptic	**5.1**°
Light-travel time from the Moon	1.3 seconds

Why are the two months different? It is the same story as the sidereal and solar day (Lecture 1.1). In 27.3 days the Moon completes one orbit relative to the stars, but meanwhile the Earth–Moon pair has moved about 27° around the Sun, so the Moon needs about 2.2 more days to line up with the Sun again.

The Moon rises about 50 minutes later every day

The Moon gains 360° on the Sun in 29.53 days:

$$\frac{360^\circ}{29.53\ \text{d}} \approx 12.2^\circ \text{ per day} \quad \Longrightarrow \quad 12.2^\circ \times 4\ \frac{\text{min}}{\text{deg}} \approx 49 \text{ minutes later each day}$$

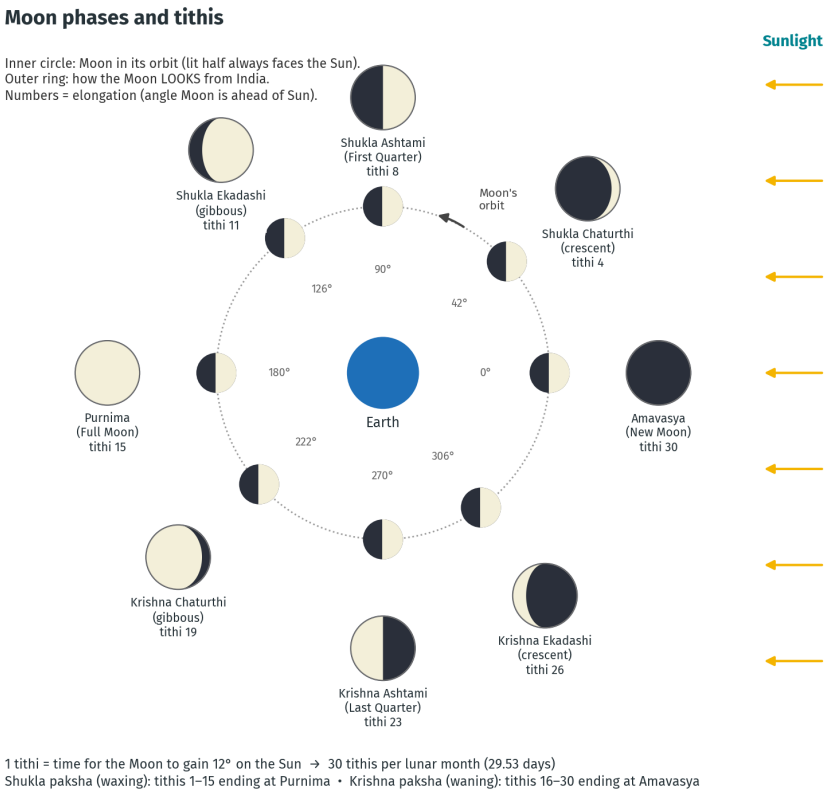
(on average; the actual delay varies from about 30 to 70 minutes).

2. Why the Moon has phases

The Moon makes no light of its own; it reflects sunlight. **Half of the Moon is always lit by the Sun.** As the Moon orbits Earth we see different amounts of that lit half. The angle between the Moon and the Sun, as seen from Earth, is called the **elongation**.

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P.T.O.



Inner circle: the Moon in its orbit (its lit half always faces the Sun). Outer ring: how the Moon looks from India.

Indian calendars count the lunar month in 30 *lunar days* called **tithis**, grouped into two fortnights: **Shukla paksha** (waxing, bright) ending at **Purnima** (Full Moon), and **Krishna paksha** (waning, dark) ending at **Amavasya** (New Moon). The names used below simply count the days: *Chaturthi* = 4th, *Ashtami* = 8th, *Ekadashi* = 11th.

Elong.	Phase	Indian name	Rises	Highest	Sets
0°	**New Moon**	Amavasya	~6 AM	~noon (invisible)	~6 PM
~ 45°	Waxing crescent	Shukla Chaturthi	~9 AM	~3 PM	~9 PM — seen in the west after sunset
90°	**First Quarter**	Shukla Ashtami	~noon	~6 PM	~midnight
~ 135°	Waxing gibbous	Shukla Ekadashi	~3 PM	~9 PM	~3 AM
180°	**Full Moon**	Purnima	**~6 PM (sunset)**	~midnight	**~6 AM (sunrise)**
~ 225°	Waning gibbous	Krishna Chaturthi	~9 PM	~3 AM	~9 AM
270°	**Last Quarter**	Krishna Ashtami	~midnight	~6 AM	~noon
~ 315°	Waning crescent	Krishna Ekadashi	~3 AM	~9 AM	~3 PM — seen in the east before dawn

Pattern to remember: the Moon rises about **6 hours later for every quarter** of its cycle ($90° \times 4$ min/deg $= 360$ min $= 6$ h).

- **Waxing** (Shukla paksha): the lit side faces **west**, towards the setting Sun.
- **Waning** (Krishna paksha): the lit side faces **east**.

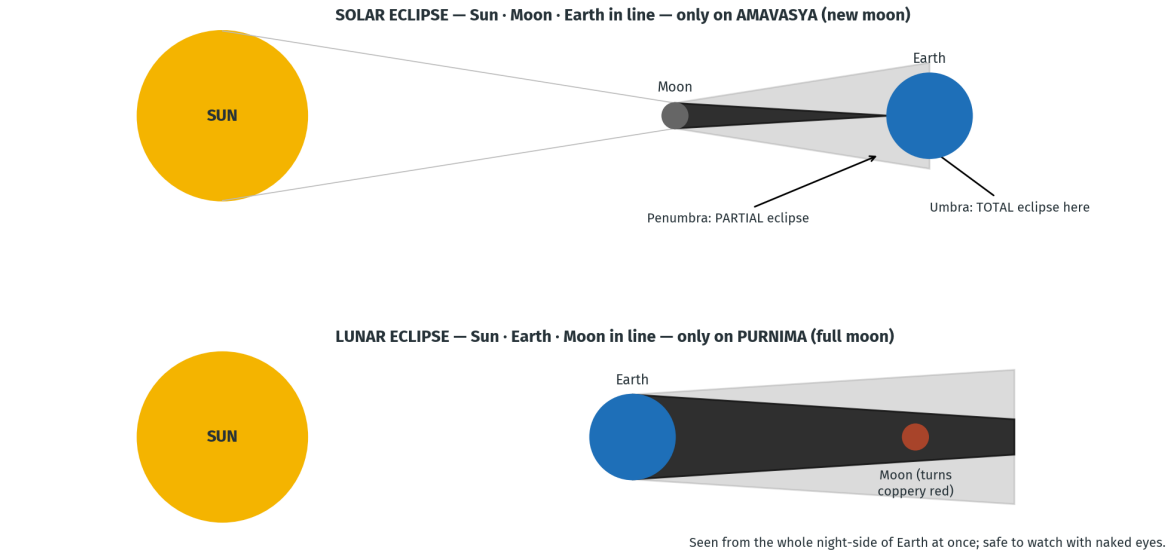
Indian sky tip

From India the young crescent often lies tilted on its back like a boat or a smile. From Europe it stands more upright, like a "D".

The Moon always shows the same face

The Moon spins once on its axis in exactly the time it takes to orbit Earth (27.3 days) — this is called **tidal locking**. Thanks to small wobbles (*libration*) we see about **59%** of its surface over time. The "far side" is not dark — it gets as much sunlight as the near side; we simply never see it from Earth.

3. Eclipses: when shadows line up



Geometry of solar and lunar eclipses (not to scale).

Solar eclipse (*Surya Grahan*)

The **Moon passes between the Sun and Earth** and its shadow falls on Earth — possible only at **New Moon (Amavasya)**.

Type	What you see	Why
Total	Sun completely covered; the sky darkens; the pearly **corona** appears	You are inside the Moon's dark shadow (umbra)
Annular	A "ring of fire"	The Moon is near its farthest point and looks slightly too small
Partial	A "bite" taken out of the Sun	You are in the outer shadow (penumbra)
Hybrid	Annular in some places, total in others	A rare in-between case



Totality during the 21 August 2017 eclipse: the Sun's glowing outer atmosphere, the corona, is visible only when the Moon covers the bright disc completely.

Photo: NASA/Carla Thomas (Armstrong Flight Research Center), public domain

A cosmic coincidence

The Sun is about 400 times wider than the Moon but also about 400 times farther away, so both look almost exactly the same size ($\approx 0.5°$) in our sky. That is why total solar eclipses are possible. Totality lasts at most about $7\frac{1}{2}$ minutes.

Safety first

Never look at the Sun directly, or through binoculars, a telescope or a camera, without a **certified solar filter**. Even when 99% covered, the Sun can permanently damage your eyes. Use ISO 12312-2 eclipse glasses, or **pinhole projection** (sunlight through a small hole — or the gaps between tree leaves — projected onto a surface).

Lunar eclipse (*Chandra Grahan*)

Earth passes between the Sun and the Moon and the Moon moves into Earth's shadow — possible only at **Full Moon** (**Purnima**).

- **Total:** the whole Moon turns coppery red — a "blood Moon".
- **Partial:** part of the Moon is in the dark umbra.
- **Penumbral:** only a slight dimming — easy to miss.

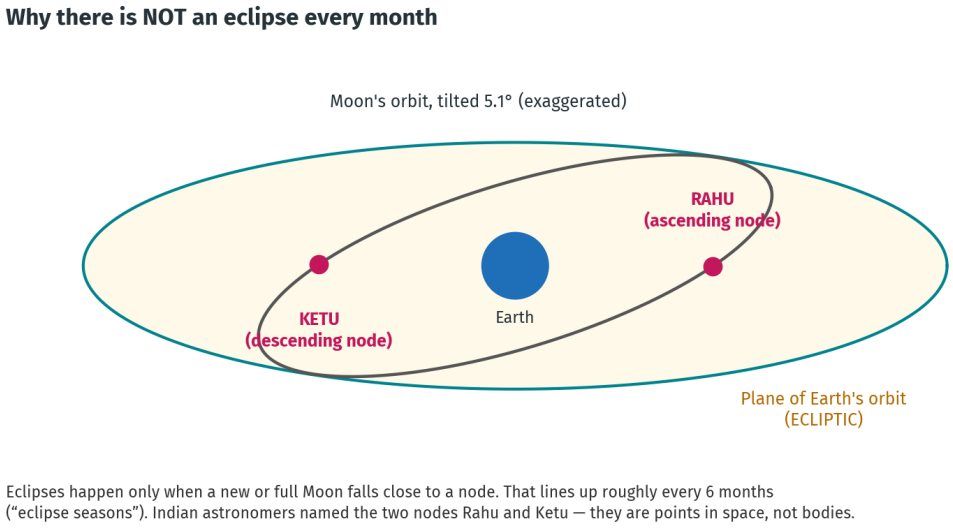
Why red? Sunlight passing through Earth's atmosphere is bent (refracted) into the shadow. Blue light is scattered away (the same reason sunsets are red), so mainly red light reaches the Moon. During totality the Moon is lit by all the sunrises and sunsets on Earth at that moment! A lunar eclipse is seen by **everyone on the night side of Earth** and is **completely safe** to watch.



The total lunar eclipse of 31 January 2018, photographed from Kozhikode (Calicut), Kerala.

Photo: Irvin Calicut, CC BY-SA 4.0, via Wikimedia Commons

4. Why not an eclipse every month? Rahu and Ketu



The Moon's orbit is tilted 5.1° to Earth's orbit; eclipses can happen only near the two crossing points (nodes).

At most New and Full Moons the Moon passes a little **above or below** the Sun–Earth line, so no shadow falls. An eclipse can occur only when a New or Full Moon happens **near one of the two points where the Moon's orbit crosses the ecliptic**, the **nodes**:

- the **ascending node** (Moon crossing northward) — **Rahu** in Indian astronomy;
- the **descending node** (Moon crossing southward) — **Ketu**.

So the mythological "demons that swallow the Sun and Moon" are, astronomically, the two **nodes** where eclipses happen. Indian astronomers tracked them precisely and counted them among the nine *grahas*, even though they are points in space, not objects.

Indian connection — Aryabhata

Aryabhata (499 CE) correctly explained that a lunar eclipse is the Moon entering Earth's shadow and a solar eclipse is the Moon covering the Sun, and gave methods to calculate their size and duration.

Eclipse seasons and the Saros

- The Sun passes a node about every 173 days, so eclipses come in **eclipse seasons** roughly **6 months apart**. Every year has **at least 2 and at most 5** solar eclipses.
- Eclipses repeat in a cycle called the **Saros**: 18 years 11 days 8 hours (6,585.3 days). The Babylonians used it to predict eclipses more than 2,500 years ago.

5. Eclipses and India — history and what's coming

Date	Eclipse	Why it matters
18 Aug 1868	Total solar, Guntur (AP)	Pierre Janssen saw a mysterious yellow line in the Sun's spectrum — later identified as a new element, **helium**, discovered from India!
24 Oct 1995	Total solar	Seen from Rajasthan (Neem ka Thana), UP and West Bengal (Diamond Harbour)
22 Jul 2009	Total solar	Longest totality of the 21st century (up to 6 m 39 s); seen from Surat, Bhopal, Varanasi, Patna, Taregna
26 Dec 2019	Annular	"Ring of fire" over Kerala, Tamil Nadu and Karnataka
21 Jun 2020	Annular	Ring of fire over Rajasthan, Haryana, Uttarakhand
7–8 Sep 2025	Total lunar	Visible across all of India

Upcoming eclipses visible from India

Date	Type	Visibility
20 Feb 2027	Penumbral lunar	subtle dimming
2 Aug 2027	Partial solar	western India, late afternoon — ~30% covered from Mumbai
12 Jan 2028	Penumbral lunar	subtle
6 Jul 2028	Partial lunar	
22 Jul 2028	Partial solar	
31 Dec 2028	**Total lunar**	fully visible across India — a New Year's Eve blood Moon!

Source: timeanddate.com eclipse listings for India. Always confirm local timings close to the date.

Try this month — the Moon Diary

Observe the Moon on as many evenings as possible for a month. Each time, record the date and time, sketch its shape, note its direction and its height (in fists), and look up the day's tithi in a calendar. You will discover the patterns in the table above by yourself.

Key points to remember

- Half of the Moon is always lit; phases depend on the Sun–Earth–Moon angle.
- Sidereal month 27.3 d; synodic month 29.5 d; the Moon rises ∼50 min later each day.
- Full Moon rises at sunset; New Moon is near the Sun and invisible.
- **Solar eclipse = New Moon (Amavasya); lunar eclipse = Full Moon (Purnima).**
- The Moon's orbit is tilted 5.1°, so eclipses happen only near the **nodes (Rahu & Ketu)**, in eclipse seasons ∼6 months apart.
- **Never look at the Sun without a proper solar filter.**

Check yourself

1. You see the Moon high in the south at sunset. What phase is it, roughly?
2. Why can a lunar eclipse never happen on Amavasya?
3. What is the Saros, and who first used it?
4. Why is the eclipsed Moon red rather than black?

Answers: 1. First Quarter (Shukla Ashtami). 2. The Moon must be opposite the Sun, behind Earth — only at Full Moon. 3. An 18 y 11 d 8 h eclipse cycle, used by the Babylonians. 4. Earth's atmosphere bends red sunlight into the shadow.

Further reading (free / public)

- OpenStax, *Astronomy 2e*, §4.5 "Phases and Motions of the Moon" and §4.7 "Eclipses of the Sun and Moon" — `https://openstax.org/books/astronomy-2e/pages/4-5-phases-and-motions-of-the-moon`
- NASA Eclipse website — `https://eclipse.gsfc.nasa.gov` and `https://www.timeanddate.com/eclipse/in/india`